

$R(A, B, C, D, E)$

$FD \Rightarrow \{ AB \rightarrow CE, E \rightarrow AB, C \rightarrow D \}$

is it in 3NF or not?

$C.K \Rightarrow AB, E.$

$AB \rightarrow CE$

- (i) Non-trivial $\rightarrow \checkmark$
- (ii) AB is Superkey $\rightarrow \checkmark$
- (iii) ~~E is a~~ $\checkmark$

Prime Attribute $\Rightarrow A, B, E.$
Non Prime $\Rightarrow C, D.$

$E \rightarrow AB$

- $\rightarrow$ (i) $\times$
- (ii) $\checkmark$
- (iii) $\checkmark$

$C \rightarrow D$

(i)	$\times$
(ii)	$\times$
(iii)	$\times$

It is not in 3NF.

3NF! $\rightarrow$ if $X \rightarrow Y$.

Then (i) $X \rightarrow Y$ must be trivial.

(ii) X is superkey

(iii) Y should be a prime Attribute.



BCNF! $\rightarrow$ (Boyce Codd Normal Form).

(i) $X \rightarrow Y$ must be trivial.
(ii) X is a superkey.

BCNF is stricter than 3NF.

BCNF may not preserve the FD.
It is preserved upto 3NF.

Multi-valued Dependency

ANF.

$$X \twoheadrightarrow Y$$

- (i) $Y \subseteq X$ OR.
- (ii) ~~$X \cup Y = R$~~

$$R(ABCD)$$
$$AB \twoheadrightarrow CD$$

$$AB \cup CD = R$$
$$AB \cup CD = R$$

$$X \twoheadrightarrow A$$

- (i) $X \twoheadrightarrow A$ must be trivial.
- (ii) X is a superkey.

$$R(ABC), F = \{ A \twoheadrightarrow BC, B \twoheadrightarrow AC \}$$

$$(i) A \twoheadrightarrow BC$$
$$A \cup BC = R$$

$$(ii) B \twoheadrightarrow AC$$
$$B \cup AC = R$$

So, It is in ANF.

Join Dependency

$$R = \{R_1, R_2, R_3\}.$$

Let, $R_1 \bowtie R_2 \neq R$, $R_2 \bowtie R_3 \neq R$,
 $R_1 \bowtie R_3 \neq R$. $R_1 \bowtie R_2 \bowtie R_3 = R$.

$\xrightarrow{\hspace{2cm}}$
 $R(A, B, C)$, $F = \{A \rightarrow BC, C \rightarrow BA\}$.
 $JD = \{ \textcircled{BC}, \textcircled{AB} \}$. is it in 5NF or not?
 $BC^+ \Rightarrow BCA$ $AB^+ \Rightarrow ABC$.

Let, $R(ABCDEI)$

$F = \{ A \rightarrow BCDE, BC \rightarrow AI \}$

$JD = \{ ABCD, CDE, BDI \}$

$ABCD^T \Rightarrow ABCDEI$

✓

$CDE^T \Rightarrow CDE$

CDE is not a superkey.

So, it is not in 5NF.

$R(ABCD)$, FD $\{ \boxed{A \rightarrow C}, B \rightarrow D \}$

☒ (A) 1NF not in 2NF.

(C) 3NF not in BCNF

$A \rightarrow A$
 $A \rightarrow A$
 $A \rightarrow A$

$AB \rightarrow ABCD$
 $\boxed{A, B}$

(B) 2NF not in 3NF

(D) None of the above.

C.D.

